

MAYUR VEER

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EDUCATION

Northeastern University, Boston, MA

Expected Dec 2026

Master of Science in Software Engineering Systems

Relevant courses: System Design in Age of AI, Network Structures and Cloud Computing, Program Structure and Algorithms, Concepts of Object-Oriented Design, Web Design and User Experience Engineering

PROFESSIONAL EXPERIENCE

Cybersenz LLC, Remote

Jan 2026 – May 2026

Software Engineering Intern

- Containerized a **6-service stack** (FastAPI, React, email agent, Postgres, Redis, Vault) into a **single-command Docker Compose deploy** with in-container Alembic migrations and auto-restart on all 6 services for self-healing recovery
- Hardened Azure production to **zero secrets on disk** by injecting **Key Vault** secrets via **Managed Identity** at startup, closing public access to Postgres and Vault, and terminating TLS at nginx
- Shipped production clinic software with **50+ JWT-secured FastAPI endpoints** and React dashboards for FHIR-style patient registration, role-based access and audit logging, integrated with **Prognosis EHR** and **HMAC-signed webhooks**

Ernst & Young LLP, Mumbai, INDIA

Sept 2021 – Aug 2024

Team Lead – Software Engineer

- Accelerated invoice processing by **17x** by reengineering data script with Pandas optimizations, enabling team to process **25,000 invoices in 3 minutes** which took 1 hour for 1,800 previously, eliminating processing bottleneck for monthly billing operations and reducing infrastructure costs by **40%**
- Delivered **90% manual effort reduction** for Formula 1 by automating Arabic invoice generation with context-aware PDF translation, processing **500+ invoices weekly** at **99.8% accuracy**
- Reduced application response time by **60%** through React optimization (useCallback, Context API, localStorage), improving UX for **100+ daily users** across modules and decreasing bounce rate by **35%**
- Cut page load latency by **50%** by implementing Angular lifecycle optimization with conditional data refresh using ngOnChanges, reducing client-side utilization by **60%**
- Led **4-person engineering team** in delivering full-stack application on schedule, establishing code review standards that reduced production bugs by **40%** and improved deployment velocity by **4x**

PROJECTS

Self-Hosted PaaS with Fault-Tolerant Scheduler | Python, FastAPI, Postgres, K8s, Terraform

- Built a Heroku-style PaaS where git push triggers build → deploy → verify pipelines with Kaniko image builds, immutable releases, and **automatic rollback** on failed health checks via a FastAPI control plane
- Engineered a distributed job scheduler from scratch using **Raft leader election** (pre-vote, check-quorum) and Postgres fencing epochs, with heartbeat leases, exponential backoff, and a dead-letter queue
- Validated fault tolerance with a custom chaos harness that kills, freezes, and partitions nodes under **25 jobs/s** load, losing **0 of 4,652 jobs** and preserving **exactly-once effects** across **236 duplicate executions**
- Instrumented Prometheus/Grafana dashboards, alert rules, and OpenTelemetry traces spanning API to every job attempt, and authored Terraform modules and customize overlays for EKS, GKE, and kind

Real-Time Collaborative Code Review Platform | Python, React, TS, TF, Docker, GCP, K8s

- Built production platform with WebSocket real-time collaboration supporting **100+ concurrent users** at **<200ms latency** using Redis pub/sub and PostgreSQL
- Achieved **99.9% uptime** via Kubernetes deployment across 3 GCP zones with autoscaling (2-10 replicas) and Prometheus/Grafana monitoring tracking **15+ SLIs**
- Automated deployment with Terraform (IaC managing **20+ GCP resources**) and GitHub Actions CI/CD featuring **50+ test cases**, achieving **90% code coverage** and **<5 min deployment cycles**
- Secured infrastructure with GCP KMS encryption, Secret Manager integration, and OAuth 2.0 authentication with role-based access control

Data Processor | Python, FastAPI, GCP, Terraform, Docker, GitHub Actions, Pub/Sub

- Architected serverless platform handling **2,133 requests/min** with **1,169ms p95 latency**, processing **3M+ monthly requests** at **60% lower cost** than VM alternative
- Deployed async Pub/Sub architecture achieving **95% success rate** and **zero message loss** through automatic NACK-based retries, dead-letter queues, and exponential backoff
- Designed multi-tenant security using Firestore sub-collections with physical data isolation and PII redaction, supporting **50+ concurrent tenants**
- Orchestrated full deployment lifecycle using Test-Driven Development, with Terraform managing **15+ GCP resources** and GitHub Actions CI/CD running **21 PyTest suites** at **85% coverage**

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, HTML, CSS, SCSS, SQL, Shell, YAML, HCL
- Frameworks & Tools:** Flask, FastAPI, React, Angular, Power BI, Tableau
- Databases:** MSSQL, PostgreSQL, SQLite, MongoDB, Firestore, Redis
- DevOps:** CI/CD pipelines, Kubernetes, Docker, GitHub Actions, Terraform, Packer, AWS, Azure, GCP, Linux